

Project Report

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Branch: MCA

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Subject Name: IOT Lab

UID: 24MCA20093

Section/Group: 24MCA 3-(A)

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Aim: IoT Based Smart Medicine Dispenser System to monitor and ensure timely intake of medicines.

Objectives:

- To automate medicine dispensing based on scheduled timings.
- To provide alerts using buzzer and LED for medicine intake.
- To monitor and control the system remotely using IoT platform.
- To implement a confirmation mechanism using a push button before dispensing medicine.
- To ensure timely and reliable medicine intake through notification alerts.

Components Required:

Sno	Name of Component	Qty.
1	NodeMCU (ESP8266)	1
2	Arduino Uno	1
3	Servo Motor	3
4	Buzzer	1
5	LED	3
6	Push Button	1
7	Connecting Wires (Jumper Wires)	As required
8	Breadboard	1

Details of Components:

1. NodeMCU:

The NodeMCU board is a low-cost, open-source IoT platform based on the ESP8266 WiFi module. It enables wireless communication and is widely used for IoT-based applications. It acts as the main controller in this project, handling all operations such as timing, communication, and device control.

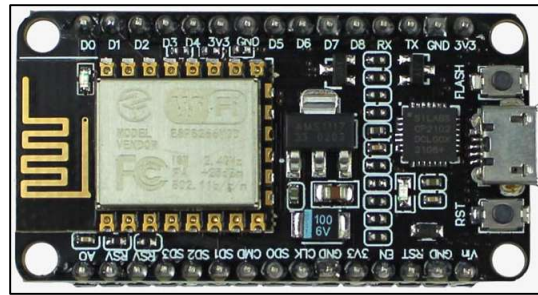


Figure 1: NodeMCU Board

2. Arduino UNO:

The Arduino UNO is used in this system to provide stable power to the servo motors. Since multiple servo motors require higher current than the NodeMCU can safely provide, the Arduino helps in maintaining stable power distribution. The main control logic is handled by the NodeMCU, while the Arduino supports the system by ensuring reliable operation of the servo motors.



Figure 2: NodeMCU Board

3. Servo Motor:

Servo motors are used to open and close the medicine boxes. Each servo is connected to a specific compartment and rotates to a fixed angle when triggered. After dispensing the medicine, the servo returns to its original position.

4. Buzzer:

The buzzer is used to alert the user when it is time to take medicine. It generates a sound signal to notify the user.

5. LED:

LEDs are used as visual indicators. Each LED represents a medicine box and glows when that particular box is activated.

6. Push Button:

The push button is used as an input device for confirmation. After the buzzer alert, the system waits for the user to press the button before dispensing the medicine.

7. Breadboard:

A breadboard is used to assemble the circuit without soldering and allows easy connection of all components.

8. Connecting Wires (Jumper Wires):

These wires are used to connect all components together and ensure proper signal and power flow in the circuit.

Block Diagram of Designed Model:

The system consists of a NodeMCU connected to servo motors, LEDs, a buzzer, and a push button. The NodeMCU communicates with the Blynk IoT platform via WiFi. The user sets the medicine schedule through the Blynk app. At the scheduled time, the NodeMCU activates the buzzer and waits for user confirmation through the push button. Upon confirmation, the corresponding servo motor rotates to open the medicine box, and the LED indicates the active box.

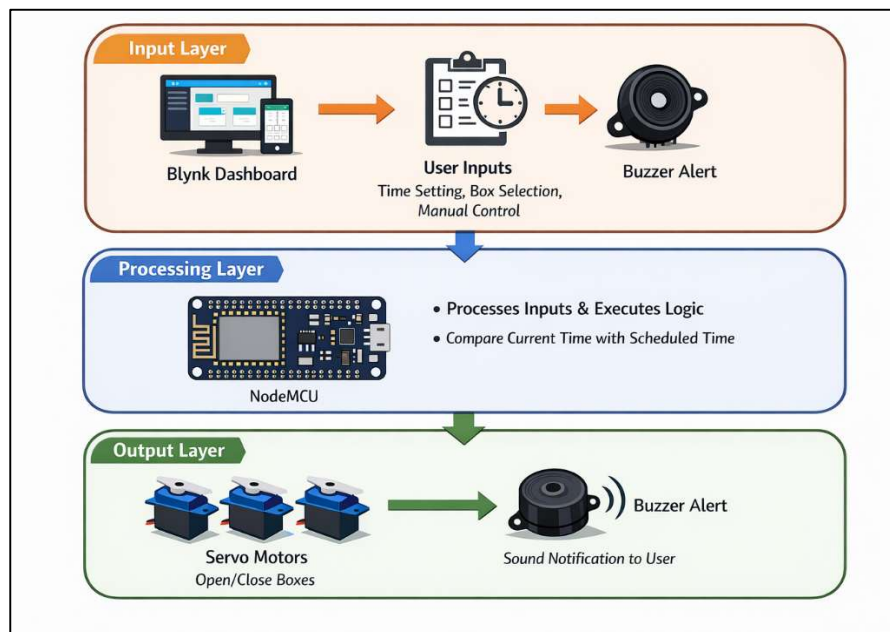


Figure 3 : Block Diagram of Model

Working of Designed Model:

- As shown in Figure 3, the Smart Medicine Dispenser is connected to a WiFi network using the NodeMCU (ESP8266), which acts as the main controller of the system. The NodeMCU communicates with the Blynk IoT platform, allowing the user to remotely set and manage medicine schedules through a mobile dashboard.
- The user sets specific timings for medicine intake such as morning, afternoon, and night. These timings are stored in the system and continuously monitored using real-time clock synchronization through the internet.

- When the current time matches the scheduled time, the system activates the buzzer to alert the user. This alert serves as a reminder that it is time to take the medicine. At the same time, the system enters a waiting state and allows the user a fixed time window (around 15 seconds) to respond.
- During this time, the user is required to press the push button to confirm that they are ready to take the medicine. If the button is pressed within the given time, the NodeMCU triggers the corresponding servo motors, which rotate to open the selected medicine boxes. Simultaneously, the LEDs glow to indicate which boxes are active.
- After dispensing, the servo motors return to their original position, and the LEDs turn off. The system also sends a notification through the Blynk platform indicating that the medicine has been successfully taken.
- If the user does not press the push button within the given time window, the system assumes that the medicine has been missed. In this case, a notification is sent to the user indicating that the dose was not taken.
- The system also supports manual control, where the user can trigger dispensing directly from the Blynk dashboard. Even in manual mode, the buzzer alert and push button confirmation mechanism remain active to ensure proper monitoring.

Pictures of Prototype:



Figure 4 : Designed Model

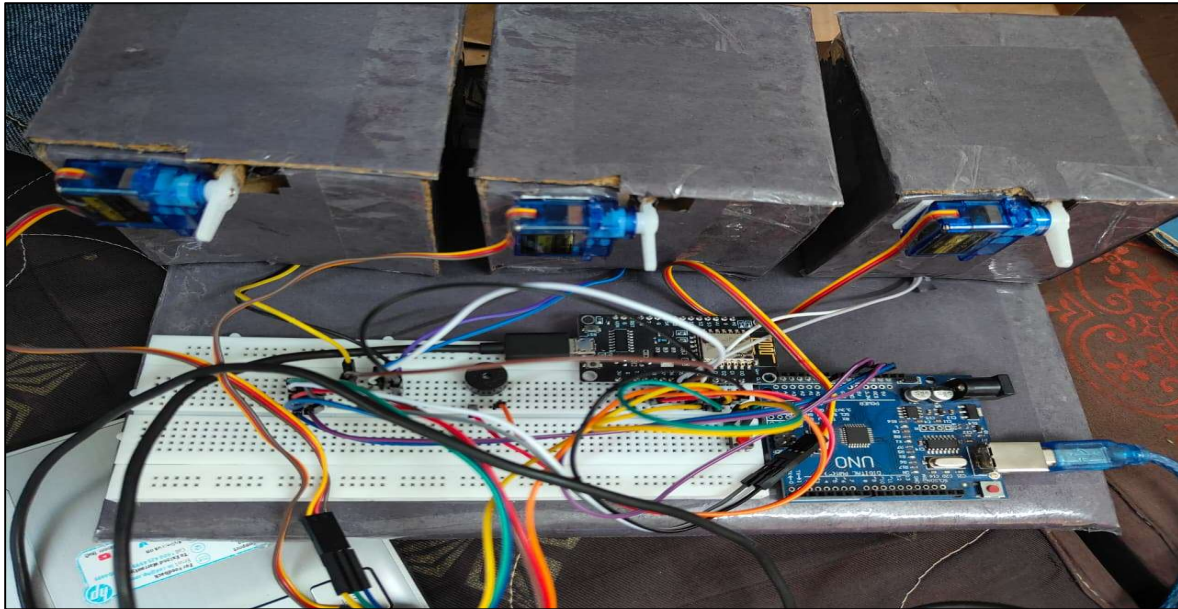


Figure 5 : Hardware Model

Output of Deigned Model/Prototype

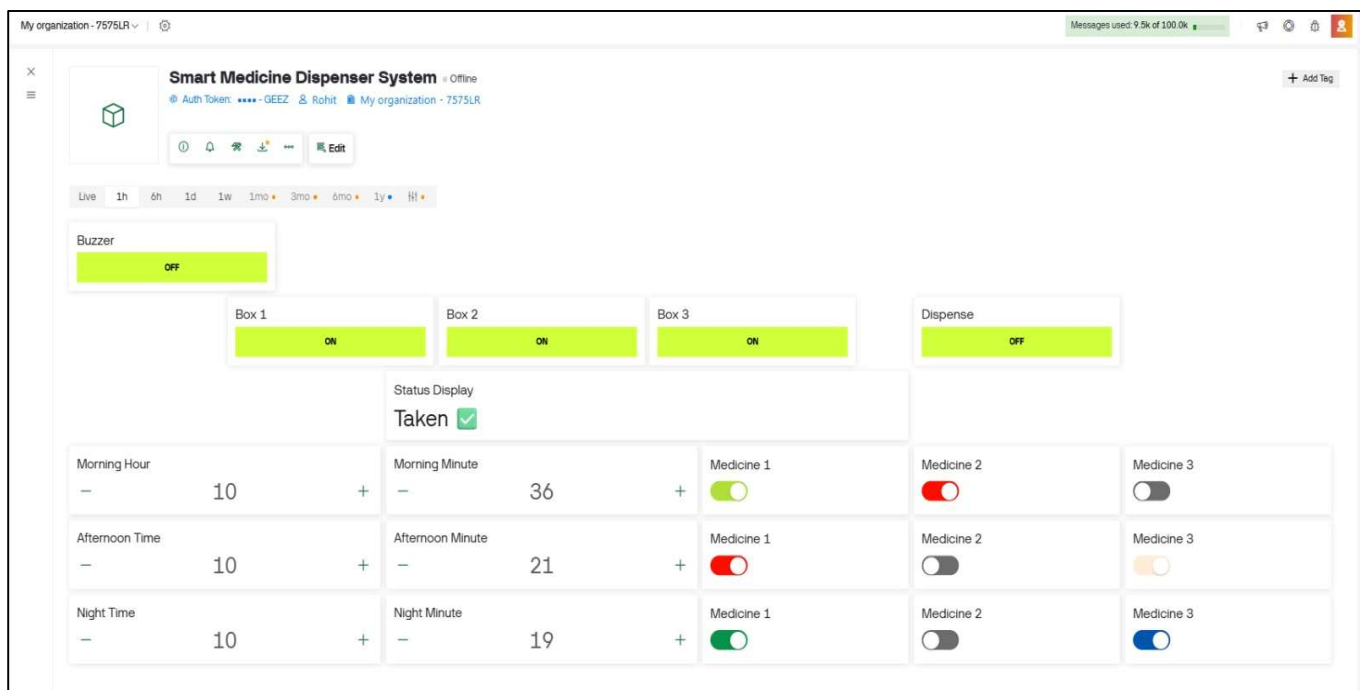


Figure 6 : Blynk Dashboard



Figure 7 : Email Notification(Medicine Not Taken)



Figure 8 : Email Notification(Medicine Taken)

Learning outcomes (What I have learnt):

1. Learned how to interface NodeMCU with sensors and actuators.
2. Understood the working of IoT platforms like Blynk.
3. Gained knowledge about automation and embedded systems.
4. Improved skills in circuit design and troubleshooting.